Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

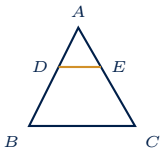
1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)

1. The shadow of a 5 m high pole is 2 m. At the same time, the shadow of a tree is 12.5 m. The height of the tree is:

(a) 31.25 m (b) 25 m (c) 28.5 m (d) 30 m

2. In the given figure, $DE \parallel BC$. If $\frac{AD}{DB} = \frac{2}{3}$ and $AC = 18$ cm, then AE is equal to:



(a) 7.2 cm (b) 6.0 cm (c) 8.4 cm (d) 5.4 cm

3. In an equilateral triangle of side $2a$, the length of each altitude is:

(a) $\sqrt{3}a$ (b) $2\sqrt{3}a$ (c) $\frac{\sqrt{3}}{2}a$ (d) $3a$

4. If $\Delta ABC \sim \Delta DEF$ with $\angle A = 45°$ and $\angle E = 55°$, then the degree measure of $\angle C$ is:

(a) 80° (b) 70° (c) 90° (d) 85°

5. If two similar triangles have their corresponding medians in the ratio 5 : 7, the ratio of their perimeters is:

(a) 25 : 49 (b) 5 : 7 (c) 7 : 5 (d) $\sqrt{5} : \sqrt{7}$

6. Assertion (A): In ΔABC, D and E are on AB, AC such that $AD = 2, AB = 6, AE = 3, AC = 9$. Then $DE \parallel BC$.
Reason (R): If $\frac{AD}{AB} = \frac{AE}{AC}$, by the converse of Basic Proportionality Theorem, $DE \parallel BC$.

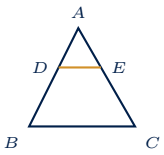
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

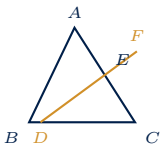
7. **[2 Marks]** A girl of height 100 cm walks away from the base of a lamp-post 4 m high at a speed of 1.5 m/s. Find the length of her shadow after 3 seconds.

8. **[2 Marks]** If $\Delta ABC \sim \Delta DEF$, $AB = 4$ cm, $DE = 6$ cm, $EF = 9$ cm, and $FD = 12$ cm, find the perimeter of ΔABC.

SECTION C: SHORT ANSWER QUESTIONS (6 Marks)

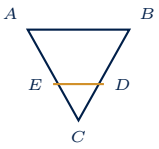


9. **[3 Marks]** In ΔABC, D and E divide AB and AC such that $AD : DB = 2 : 3$. If $BC = 15$ cm, calculate the length of DE.

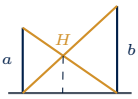


10. **[3 Marks]** In the given figure, line segment DF intersects side AC of ΔABC at E such that E is the midpoint of CA and $\angle AEF = \angle C$. Prove that $\Delta AEF \sim \Delta CDE$.

SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)



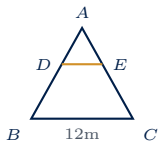
11. **[5 Marks]** A surveyor wishes to find the distance between two inaccessible landmarks A and B across a river. Setting up sightlines from accessible bank station C with points E on CA and D on CB such that $ED \parallel AB$, prove that $\Delta CED \sim \Delta CAB$. If $CE = 10$ m, $CA = 40$ m, and $ED = 25$ m, calculate distance AB.



12. **[5 Marks]** Two vertical poles of heights a and b stand on level ground. Ropes run from the top of each pole to the foot of the other. Prove that the height H of their intersection point above the ground is $\frac{ab}{a+b}$, and prove that H is completely independent of the distance between the poles.

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 12: Solar Photovoltaic Truss & Triangular Gable Support Structure**



A solar power installation requires a triangular steel roof truss ABC with $AB = AC = 10$ m and base $BC = 12$ m. A horizontal reinforcing tie-rod DE is welded parallel to base BC at a height such that D and E lie on AB and AC with $AD = 4$ m. Solar photovoltaic panels are mounted along the rafters.

(i) **[1 Mark]** What is the length of segment DB along the rafter?

(ii) **[1 Mark]** State the similarity criterion between ΔADE and ΔABC.

(iii) **[2 Marks]** Calculate the length of the reinforcing tie-rod DE and the perimeter of the upper truss ΔADE.

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★